**AGENDA:**

1. **Ratings of each paper**
2. **Verdict on which topic to work**
3. **Which topics to study to gather knowledge**
4. **What kind of softwares to use and their skill levels**
5. **Fix an assignment topic to give to our teammates to check their level of commitment.**

**Table of Contents:**

# List of papers reviewed

a) A MOSFET SPICE Model With Integrated Electro-Thermal Averaged Modeling, Aging, and Lifetime Estimation, doi: 10.1109/ACCESS.2020.3047110
b) AQuasi-Floating Channel-Based SPICE Model for Improving the Modeling Accuracy of SiC MOSFETs With Multiple Device Structures, doi: 10.1109/TPEL.2024.3433370
c) JFET and MOSFET SPICE Models in a Wide Temperature Range, DOI: 10.1134/S1063739721070064
d) Physics-Based PiN Diode SPICE Model for Power-Circuit Simulation, doi: 10.1109/TIA.2007.900492
e) A Flexible, Physically based SPICE Model for the JFET Resistance in Silicon Carbide and Super-Junction MOSFETs
f) Electro-Thermal Modeling of SiC Power Devices for Circuit Simulation, doi: 10.1109/IECON.2013.6699223.
g) RCFilter Design for Wireless Power Transfer: AFourier Series Approach, doi: 10.1109/LSP.2022.3148673
h) SPICE Analysis of RL and RC Snubber Circuits for Synchronous Buck DC-DC Converters, doi: 10.1109/MIPRO.2015.7160245
i) High Precision I-V Characteristics SPICE Model for Silicon Carbide MOSFET, DOI: 10.1109

# Analysis of the Research Papers

**Topic Categories**

1. **Power Electronics**: The papers address power MOSFET technologies, specifically SiC and Super-Junction MOSFETs, which are critical for efficient power conversion.
2. **Semiconductor Device Modeling**: Few papers propose a physically based SPICE model to simulate the JFET region resistance, accounting for nonlinear effects like junction depletion and velocity saturation.
3. **Circuit Simulation**: The model is designed to be SPICE-agnostic, enabling accurate simulation of conduction losses and other figures of merit like Short Circuit Withstand Time.
4. **Material Science**: The paper discusses SiC, a wide-bandgap material, and its advantages over silicon in power applications.

**Problems Addressed**

- **Inaccurate Modeling of JFET Region Resistance**: Existing models use simple linear resistors, failing to capture nonlinear effects in the JFET region, which contributes significantly to total on-resistance (25% for SiC MOSFETs, 69% for Super-Junction MOSFETs).
- **Conduction Loss Simulation**: Accurate modeling of JFET resistance is crucial for simulating conduction losses in power electronic applications.
- **Nonlinear Effects in Drift Region**: Previous models have limitations in capturing nonlinear effects like junction depletion and velocity saturation, impacting simulation accuracy for high-voltage and high-current scenarios.
- **Application-Specific Performance**: The model addresses critical metrics like Short Circuit Withstand Time, essential for applications such as traction inverters.

# Table of Resources and Knowledge Pools

| Resource (Software) | Description | Knowledge Pools Required |
|---|---|---|
| **SPICE (e.g., LTspice, PSpice)** | Circuit simulation software for modeling MOSFET behavior and validating the proposed JFET resistance model. | Semiconductor physics, circuit analysis, SPICE modeling, MOSFET operation principles. |
| **MATLAB/Simulink** | Used for numerical analysis and validation of model equations, particularly for nonlinear effects. | Numerical methods, semiconductor device physics, data analysis, control systems. |
| **TCAD (e.g., Silvaco, Synopsys)** | Technology Computer-Aided Design for simulating semiconductor device structures and validating physical models. | Semiconductor fabrication, device physics, TCAD simulation techniques, material science (SiC). |
| **Cadence Virtuoso** | Potential use for integrated circuit design and SPICE model integration. | IC design, SPICE model implementation, analog circuit simulation. |

# Topic Selection

Based on the criteria (completion within 6 months, software simulation feasibility, and available knowledge), the topic of **A Quasi-Floating Channel-Based SPICE Model for Improving the Modeling Accuracy of SiC MOSFETs With Multiple Device Structures** is recommended. This choice is justified because:

1. **Completion within 6 Months**: The research focuses on specific components and leverages existing SPICE frameworks, making it achievable with focused effort.
2. **Software Simulation Feasibility**: SPICE and MATLAB are accessible and widely used for circuit and numerical simulations, respectively. TCAD can be optional for deeper physical validation.
3. **Available Knowledge**: Semiconductor physics, MOSFET operation, and SPICE modeling are well-documented fields, with ample resources available for learning.

# Roadmap for Research

**Knowledge Fields to Study**

1. **Semiconductor Physics**:
    - **Key Concepts**: Understand carrier transport, junction depletion, velocity saturation, and wide-bandgap materials (e.g., SiC). Study how these affect JFET region resistance.
    - **Resources**: "Semiconductor Device Physics" by Sze and Ng, online courses (e.g., Coursera's "Introduction to Power Electronics").
    - **Timeline**: 1 month for basics, focusing on MOSFET operation and SiC properties.
2. **Power Electronics**:
    - **Key Concepts**: Learn about power MOSFET applications, conduction losses, and figures of merit like RDS(on) and Short Circuit Withstand Time.
    - **Resources**: "Fundamentals of Power Electronics" by Erickson and Maksimović, IEEE Power Electronics Society webinars.
    - **Timeline**: 1 month, parallel with semiconductor physics, focusing on MOSFET applications.
3. **SPICE Modeling**:
    - **Key Concepts**: Master SPICE syntax, subcircuit modeling, and parameter extraction for MOSFETs. Understand how to implement nonlinear models.

- **Resources**: LTspice tutorials, "SPICE for Power Electronics and Electric Power" by Rashid.
- **Timeline**: 1.5 months, including hands-on practice with SPICE simulations.

4. **Numerical Analysis (MATLAB)**:
    - **Key Concepts**: Use MATLAB for solving equations related to JFET current and validating model accuracy against experimental data.
    - **Resources**: MATLAB documentation, "Numerical Methods for Engineers" by Chapra and Canale.
    - **Timeline**: 1 month, overlapping with SPICE modeling, focusing on scripting and data analysis.

# Software to Learn

To research the topic "A Quasi-Floating Channel-Based SPICE Model for Improving the Modeling Accuracy of SiC MOSFETs with Multiple Device Structures," the following software tools are essential for developing, simulating, and validating the proposed SPICE model:

1. **LTspice**:
    - **Purpose**: Simulate the quasi-floating channel-based SPICE model for SiC MOSFETs, capturing the behavior of multiple device structures and their electrical characteristics.
    - **Learning Path**: Begin with LTspice tutorials on YouTube (e.g., Analog Devices channel) to understand netlist creation and simulation. Practice building MOSFET subcircuits and simulating I-V and C-V characteristics.
    - **Timeline**: 1 month to achieve proficiency in creating and simulating complex SPICE models.
    - **Resources**: LTspice official website, user forums, and "SPICE for Power Electronics and Electric Power" by Rashid.
2. **MATLAB**:
    - **Purpose**: Analyze and validate the SPICE model by fitting simulation data to theoretical equations, particularly for quasi-floating channel effects and device parameter extraction.
    - **Learning Path**: Complete MATLAB Onramp (free course on MathWorks) and practice scripting for semiconductor equations. Use MATLAB to model channel resistance and capacitance behavior.
    - **Timeline**: 1 month for basic proficiency, focusing on numerical analysis and data fitting.

- **Resources**: MathWorks tutorials, MATLAB Central community.

3. **TCAD (Optional, e.g., Silvaco or Synopsys)**:
    - **Purpose**: Simulate physical SiC MOSFET device structures to validate the quasi-floating channel model parameters, ensuring physical accuracy.
    - **Learning Path**: Study TCAD tutorials focusing on SiC MOSFET simulations, including doping profiles and channel effects.
    - **Timeline**: 1 month for basic understanding, if pursued.
    - **Resources**: Silvaco or Synopsys documentation, university access to TCAD tools.